Project (150 pts)

MATH 240 — Introduction to Statistics, Fall 2026

Background

You will be working in groups of 2-3 to perform a statistical study on a topic and dataset of your choice.

By the end of this project, you will be able to do the following:

- Connect concepts, formulas, considerations, and methods from class to answer a real-world statistical question.
- Perform various procedures in this statistical study, including but not limited to, exploratory data analysis, hypothesis testing, and regression.
- Analyze and report on findings using clear and concise language through a report and presentation.

Your group will accomplish all of the aforementioned tasks. The project will contain the following parts:

1. **Proposal**: 5 pts, due Friday, 10/30 at 11:59pm
    - Consists of forming groups and writing a project proposal.
2. **Progress Checkpoint**: 15 pts, due Friday, 11/20 at 11:59pm
    - A (very) rough draft of your written report
    - A 1 page summary of what has been accomplished and what will need to be done over the next few weeks
3. **Presentations**: 40 pts, given in lecture on Monday, 12/7
    - A *4-5 minute presentation* in-class discussing the main points of your project.
4. **Final Portfolio**: 90 pts, due Wednesday, 12/9 at 11:59pm
    - A *3-6 page written report* containing all requirements in APA format (e.g. 12-point font, Times New Roman, double-spaced).
    - A script/Quarto document of R code showing all models and analysis done. This should be reproducible.

Breakdown of Project

Your group will be tasked with conducting a statistical study motivated by one of the following datasets seen in the `openintro` package. You can access a list of these datasets here: https://www.openintro.org/data/. Additionally, you can also find a dataset of your own online (e.g. through Kaggle, UCI Machine Learning Repository, Sports Reference). However, all datasets must be approved by me going into the proposal.

Your group will then develop a research question stemming from the topic of the dataset, which should provide an opportunity to perform multiple statistical methods on your data. For instance, if I am working with basketball data, perhaps I ask, "what is the impact of turnovers and fouls on a player's minutes throughout the season?" In this question, I can run an exploratory data analysis, a linear regression fit, and run some hypothesis testing in various manners.

This research question, along with a discussion of,

- the dataset you are using,
- the characteristics of the data (e.g. missing data, lack/abundance of variables, size), and
- a loose plan for investigating the question...

will be written up in the **project proposal**. More details about formatting can be found in the Components section of this document.

Once the proposal is approved by me, your group will need to run through these tasks:

1. Exploratory data analysis
    - e.g. plots and tables, initial inference
2. Main analysis
    - e.g. a tuned regression model, hypothesis testing of a mean or group of means, resampling or simulation of a probability model
3. Secondary analyses
    - e.g. use of transformed variables, looking at particular subpopulations

For the following three tasks, you will need to do the following:

1. **Exploratory data analysis**: This should involve an investigation of all variables you plan to use in your later analyses. To that end, you must provide:
    - Appropriate figures of either the variables by themselves or the relationships between variables.
        - For instance, you might be more compelled to look at certain variables one-at-a-time, but across groups. Or you may be curious about the association between any two continuous variables.
        - These figures should be labeled properly and look clean.
    - Tables of metrics/counts when necessary to convey your points.
        - You may want to point out the counts across two categorical variables. You might also want to show five-number summaries of a numerical variable between multiple groups.
    - Written analyses of the behavior within the plots and in the data.
        - Includes a discussion of visual observations, metrics, and relation to the situational context.
2. **Main analysis**: Pick between the following analyses:
    - Regression model with use of at least three predictors.
        - Initial model, potential variable selection and transformations, interpretation, and predictions.
    - Hypothesis test for a proportion, mean, or difference of means.
        - Identifying H_0/H_1, verifying conditions, running test, conclusions.
    - Resampling of the data to produce confidence intervals of particular metrics.
        - Bootstrapping, calculating central tendency metrics with confidence intervals, interpretations.

    You may also go beyond the steps listed or pivot in a different direction. Let me know before doing so. Regardless of which direction you pick, you will need to perform the model in R and discuss the consequences of your results.
3. **Secondary analyses**: These secondary analyses can be tied to your main analysis or an extension of your exploratory data analysis. For instance, you might improve your model or test for a different variable as a secondary analysis. You might instead evaluate two different groups with regard to various variables. Similar to the main analysis, you will perform this in R and discuss consequences of your secondary analysis.

As mentioned here and there, you are expected to run these analyses in R using code discussed in class and in the textbook. The results from the code (such as plots and tables) should be included in your report. The code itself will be turned in separately

and must be replicable, meaning that I can run your analyses on my end with the same results produced.

Everything up to this point will be discussed in both the report and presentation. More on this in the Components section.

Note about AI

You are allowed to use AI to aid in your investigation of your research question. This may be seen in areas such as:

- Interpretation of visuals
- Code debugging
- Formatting of plots

While you are allowed to use AI, a good chunk of your grade will be your ability to personally interpret and explain what is going on in all facets of your project. A cookie-cutter response can be provided to much of your work, but I care more about what YOU think about the investigation and how YOU interpret the findings.

Components

***1. Proposal* (*5 pts; due Friday, 10/30 at 11:59pm on Gradescope*)**

For this component, you will submit a 1 page document (Times New Roman, double-spaced, without APA formatting) containing the following:

- (1 pt) A list of the names of all group members.
  - There must be either 2 or 3 group members. You may not work alone or groups more than 3.
- (2 pts) Introduce the dataset and research question you all are investigating, as well as any relevant background to these items.
- (2 pts) A plan regarding what you plan to investigate, what analyses you will run, and how you will present these results.

Have only one person in your group turn in this plan. Comments to your group's plan will be given through Gradescope once it is submitted. Proposals not submitted by the deadline will only count as a 0.5^d proportion of the max proposal grade, where d is the number of days it is late by.

2. Progress Checkpoint* (*15 pts; due Friday, 11/20 at 11:59pm on Gradescope

This progress checkpoint can be treated as a super rough draft of the final report you will turn in at the end of the semester. Quite simply, this should just be a rough version that is at least one page long consisting of a mixture of paragraphs, plots, tables, and formatting related to your investigation. You will preface this rough draft with a summary of items/tasks your group has already done, and what your group is planning to do over the next few weeks. As such, I will be looking for the following:

- (5 pts) A 1 page summary of the following:
  - What your group has done
  - What your group still needs to do
  - Any difficulties or questions your group has up to this point
- (10 pts) Evidence of progress, consisting of
  - Unfinished figures and tables
  - Some paragraphs relating to at least one of the three analyses
  - Code that has been completed

3. Presentations (40 pts; given during lecture on Monday, 12/8 from 12-12:50pm)

Your group will be presenting during lecture on Monday, December 8th, with the order to be determined the week prior.

One person from your group will need to turn in slides to Gradescope by Monday, 12/8 at 9:00am. Failure to do so will result in a 50% penalty on the slides and the presentation.

Your slide deck should be about 5-8 slides long, containing the following:

- Introduction slides: Include title, names on first slide; showcase data/variables that are being explored; give background
- Methodology/EDA slides: Showcase methodology and EDA (with diagrams/charts if needed)
- Main analysis slides: Introduce main analysis with plots and discussion
- Secondary analysis slides: Introduce secondary analysis with plots and discussion
- Conclusion slides: Present conclusions and provide summaries of the analyses

Slide content, along with what is mentioned verbally, will play into the grade your group receives for the presentation. As such, you will be graded on the following:

- (10 pts) Delivery: points mentioned clearly and succinctly; done so in an organized manner
- (10 pts) Content: slides arranged well; content is accurate and contains main points; presentation goes through logical and clear order
- (10 pts) Intuition: content delivered with a personable and strong connection to topic; grasps details of their investigation without any aids
- (10 pts) Participation: each member has some role in presentation; presented within 4-6 minutes

Ultimately, the goal is to condense your project into 5-8 slides (that will contain the main points) and present this information in a clear and concise manner. You are free to use any slide platform (e.g. Slides, Canva, PowerPoint, Prezi, Beamer); just make sure to turn in the .pdf version of these slides to Gradescope.

4. Final Portfolio (90 pts; due Wednesday, 12/9 at 11:59pm on Gradescope)

The final portfolio will consist of three items: a final report, your dataset .csv file, and all code associated with the project. This will all be put in a zipped folder, which you will turn into Gradescope.

The project report should summarize all of the work completed. Formatting-wise, your report should:

- (2.5 pts) Contain 3-6 pages (with no abstract or table of contents required; title-page not included in page count)
- (2.5 pts) Adhere to all requirements in APA format (e.g. 12-point font, Times New Roman, double-spaced) and is organized into multiple, sensible sections (with headers indicating each section)

The report should have the following parts:

- (7.5 pts) **Introduction:** An overview of the research question, the dataset you're using, characteristics of the dataset, and your plan.
- (15 pts) **Exploratory data analysis:** An exploratory investigation of all variables used for your other analyses. This should include the following components:
  - At least 2 comprehensive figures of the relationships between variables
  - At least 1 table of metrics/counts
  - Analyses of all figures and tables

  Note that your EDA should provide a comprehensive snapshot of your most important variables rather than a partial one.
- (25 pts) **Main analysis:** A discussion of which analysis was performed and how it was completed, showing the results for the analysis. This also includes a discussion of the results and their impacts to your research question and beyond.
- (15 pts) **Secondary analyses:** A discussion of a secondary analysis and how it was completed, showing the results. This also includes a discussion of the results and the impacts to your research question and beyond.
- (7.5 pts) **Conclusion:** This should act as both a reflection and summary of the project. It should include:
  - A brief, final discussion of the overall results and their impacts
  - A quick reflection on the models and process, including improvements to faults in the project

Please put your dataset's .csv file into the zipped folder. **5 points** will be deducted if the .csv file is not in the zipped folder.

You will also include R code via one or multiple Quarto documents, showcasing all of the analyses you have done.

I will be looking for the following items:

- (15 pts) Appropriate code containing data input, EDA, and analyses
  - Organized?
  - Replicable?
  - Complete?

The code should follow a logical order with some use of comments/sentences that indicate what processes are being done. No late work will be accepted here.